



UNIVERSITY OF THE PUNJAB

Eight Semester 2013
Examination B.A. 4 Year Programme

Page No.

PAPER: Multivariate Analysis

Course Code: 31A1-411

TIME ALLOWED: Three & a Half Hours

MARK: 100/100

Attempt this Paper on Separate Answer Sheet provided

SHORT QUESTIONS

(5 marks each)

Q.No.2. Define the following:

- Factor Rotations ✓
- Discriminant Analysis
- Canonical Correlations Analysis
- Principal Component Analysis

SUBJECTIVE

Q.No.3. Data for two variables give the summary

$$n = 42, \bar{x} = \begin{bmatrix} 0.562 \\ 0.691 \end{bmatrix}, s^2 = \begin{bmatrix} 0.0134 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix} \quad 238 \checkmark \checkmark$$

Test the Hypothesis $H_0: \rho = [0.562 \quad 0.691]$ for the correlation coefficient of population consisting of all values (μ_1, μ_2) . Also draw the figure.

Q.No.4. Let X_1 and X_2 be two random variables with covariance matrix

$$\Sigma = \begin{bmatrix} 9 & 16 \\ 16 & 4 \end{bmatrix}$$

- Obtain the Principal Components and find the percentage of variation explained by each
- Change Σ into correlation matrix ρ . Find the Principal components using ρ .
- Are Principal Components from (i) and (ii) same? If yes, why?

Q.No.5. The following mean vector and covariance matrix are based on a sample of 100 observations (10)

$$X_1 = \begin{bmatrix} 6.213 \\ 1.133 \end{bmatrix}, X_2 = \begin{bmatrix} 7.412 \\ 5.321 \end{bmatrix}, S_1 = \begin{bmatrix} 1.313 & 0.124 \\ 0.124 & 0.012 \end{bmatrix}, S_2 = \begin{bmatrix} 1.456 & 0.178 \\ 0.178 & 0.015 \end{bmatrix} \quad 374 \checkmark$$

Find the Fisher's Linear Discriminant function and find constant rule. Also allocate the new observations $X' = [7.2 \quad 3.1]$ to any of these two populations.

Q.No.6. A class of 140 tenth grade children received two test two open books X_1, X_2 and two closed books X_3, X_4 . The mean vector and joint covariance matrix is given below:

$$\mu = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \text{ and } \Sigma = \begin{bmatrix} 2 & 2 & 3 & 1 \\ 2 & 5 & -1 & 3 \\ 3 & -1 & 6 & -2 \\ 1 & 3 & -2 & 7 \end{bmatrix}$$

Find the canonical correlations between X_1 & X_2 and the first pair of canonical variables. Also compute the correlations between the first pair of canonical variables and their component variables.

(08)



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Multivariate Analysis
Course Code: STAT-414

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SHORT QUESTIONS

Q2. Differentiate between the following terms:

(4.4.4.4.4)

- Principal Component Analysis and Factor Analysis
- Communalities and Factor Loadings
- Common Factors and Specific Factors
- Discriminant Analysis and Canonical Correlations
- Mahalanobis Distances and Cook's Distances

SUBJECTIVE

Q3. Derive the distribution of Hotelling T^2 -statistic for testing $H_0: \mu = \mu_0$.

(08)

Q4. Consider the sample correlation matrix based on 150 observations:

(07)

$$R = \begin{bmatrix} 1.00 & 0.75 & 0.65 & 0.66 \\ & 1.00 & 0.69 & 0.73 \\ & & 1.00 & 0.67 \\ & & & 1.00 \end{bmatrix} \text{ Test the hypothesis that: } H_0: \rho = \begin{bmatrix} 1 & \rho & \rho & \rho \\ & 1 & \rho & \rho \\ & & 1 & \rho \\ & & & 1 \end{bmatrix}$$

Q5. The following mean vector and covariance matrices are based on $n_1 = n_2 = 100$ observations:

$$\bar{X}_1 = \begin{bmatrix} 6.213 \\ 3.133 \end{bmatrix}, \bar{X}_2 = \begin{bmatrix} 7.412 \\ 5.321 \end{bmatrix}, S_1 = \begin{bmatrix} 1.813 & 0.321 \\ & 0.937 \end{bmatrix}, S_2 = \begin{bmatrix} 2.193 & 1.654 \\ & 3.789 \end{bmatrix}$$

Find the Fisher's Linear Discriminant function and Discriminant rule. Also allocate the new observations $\bar{X}' = [7.2 \ 3.1]$ to any of these populations.

(08)

Q6. A class of 140 tenth grade children received four tests, two open books $X'_1 = [X_1 \ X_2]$ and two

closed books $X'_2 = [X_3 \ X_4]$. The mean vector and joint covariance matrix is given below:

$$\mu = \begin{bmatrix} -3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \text{ and } \Sigma = \begin{bmatrix} 8 & 2 & 3 & 1 \\ & 5 & -1 & 3 \\ & & 6 & -2 \\ & & & 7 \end{bmatrix}$$

Find the canonical correlations between X_1 & X_2 and the first pair of canonical variables.

(07)



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2018

Examination: B.S. 4 Years Program

Roll No. 010606

PAPER: Multivariate Analysis

Course Code: STAT-414 Part - II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q2. Define the following:

(5 marks each)

- i) Factor Rotation 340
- ii) Discriminant Analysis 362
- iii) Canonical Correlations Analysis 343
- iv) Principal Component Analysis 267

SUBJECTIVE

Q3. Data for two variables give the summary:

(06)

$$n = 42, \bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}, S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

Test the hypothesis $H_0: \mu = [0.562 \ 0.589]$. Find 95% confidence ellipse for μ consisting of all values (μ_1, μ_2) . Also draw the ellipse.

Q4. Let X_1 and X_2 be two random variables with covariance matrix:

(06)

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

- i) Obtain the Principal Components and find the percentage of variation explained by each.
- ii) Change Σ into correlation matrix P and find Principal components using P .
- iii) Are Principal components from (i) and (ii) same? If yes, why?

Q5. Consider $n_1 = n_2 = 50$ observations on two species of iris gave the following mean vectors and covariance matrices:

$$\bar{X}_1 = \begin{bmatrix} 5.006 \\ 3.428 \end{bmatrix}, \bar{X}_2 = \begin{bmatrix} 5.936 \\ 2.770 \end{bmatrix}, S_1 = \begin{bmatrix} 0.1218 & 0.0977 \\ 0.0977 & 0.1408 \end{bmatrix}, S_2 = \begin{bmatrix} 0.2611 & 0.0835 \\ 0.0835 & 0.0965 \end{bmatrix}$$

Find the Discriminant function and allocate the new observations $X' = [5.296 \ 3.213]$ to any of these species.

(06)

Q6. Derive the canonical correlations and canonical variables.

(12)

$$\underline{z} = \frac{(n_1 - 1)\underline{z}_1 + (n_2 - 1)\underline{z}_2}{n_1 + n_2 - 2}$$

$$\sqrt{\frac{n_1 + n_2}{n_1 n_2}} \frac{(\underline{z}_1 - \bar{z}_1)'}{\sqrt{S_1}} = \frac{(\underline{z}_2 - \bar{z}_2)'}{\sqrt{S_2}}$$



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

SHORT QUESTIONS

Q2. Define the following: (5 marks each)

- i) Common Factors and Specific Factors
- ii) Fisher's Discrimination Rule
- iii) Canonical Correlations Analysis
- iv) Principal Component Analysis for Standardized Variables

SUBJECTIVE

Q3. Derive the distribution of Hotelling T^2 -statistic for testing $H_0: \mu = \mu_0$. (10)

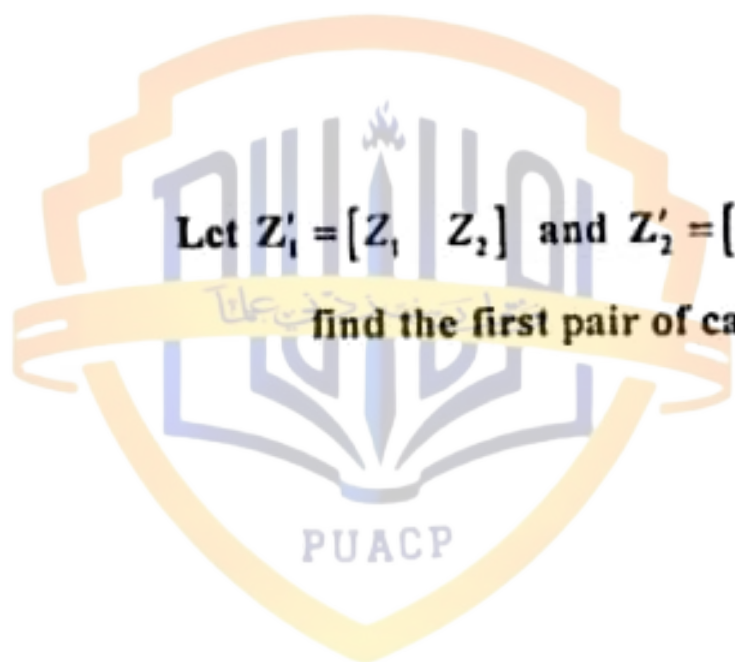
Q4. Discuss the principal components from equal correlation matrix $\rho = \begin{bmatrix} 1 & 0.6 & 0.6 \\ & 1 & 0.6 \\ & & 1 \end{bmatrix}$ (06)

Q5. Suppose that X is a multinomial random variable which comes either from P_1 with multinomial probabilities $\alpha_1, \alpha_2, \dots, \alpha_k$ or from P_2 with multinomial probabilities $\beta_1, \beta_2, \dots, \beta_k$ where $\sum_{i=1}^k \alpha_i = \sum_{i=1}^k \beta_i = 1$ and $\sum_{i=1}^k x_i = n$. Discuss the maximum likelihood discriminant rule. (06)

Q6. The covariance matrix for four standardized variables Z_1, Z_2, Z_3, Z_4 is,

$$\rho = \begin{bmatrix} 1 & 0.4 & 0.5 & 0.6 \\ & 1 & 0.3 & 0.4 \\ & & 1 & 0.2 \\ & & & 1 \end{bmatrix}$$

Let $Z'_1 = [Z_1 \ Z_2]$ and $Z'_2 = [Z_3 \ Z_4]$. Find canonical correlation between Z'_1 & Z'_2 . Also find the first pair of canonical variates. (08)





ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Solve the following questions.

(4x5=20)

Q2. a) Let $x = [X_1 \ X_2 \ \dots \ X_p]^T$ has a covariance matrix Σ with eigen values and eigen vector pairs (λ_i, e_i) . The i -th principal component is $Y_i = e_i^T x$ where $i = 1, 2, \dots, p$. Derive the correlation between Y_i and X_k . (5)

b) Show that the total variance explained by Principal Components is equivalent to the total variance of the data. (5)

For (c) and (d), the eigen-values and eigen-vectors of a correlation matrix are given below.

$$\lambda_1 = 2.85, e_1 = [0.33 \ 0.46 \ 0.38 \ 0.56 \ 0.47]^T$$

$$\lambda_2 = 1.81, e_2 = [-0.61 \ 0.39 \ -0.56 \ 0.08 \ 0.40]^T$$

$$\lambda_3 = 0.20, e_3 = [0.10 \ 0.74 \ 0.17 \ -0.60 \ -0.22]^T$$

$$\lambda_4 = 0.10, e_4 = [0.14 \ -0.28 \ 0.12 \ -0.57 \ 0.75]^T$$

$$\lambda_5 = 0.03, e_5 = [0.70 \ 0.07 \ -0.71 \ 0.00 \ 0.01]^T$$

c) Assuming two-factor model, calculate common factor loadings. (5)

d) Assuming two-factor model, calculate communalities and specific variances. (5)

Answer the following questions.

(2x15=30)

Q3. Suppose the random variables X_1, X_2 and X_3 have the covariance matrix,

$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. Derive all three Principal Components. Also, identify the percentage of variation that is explained by each Principal Component. (15)

Q4. Measurements of stiffness and bending strength (in 100 pounds per inches square) for a sample of 7 pieces of a particular grade of lumber are given below. It is known that the sample of stiffness (X_1) and bending strength (X_2) comes from bivariate normal distribution. (15)

X_1	12	11	22	18	19	16	15
X_2	41	66	76	109	108	76	69

Obtain and interpret 95% confidence ellipse for the population mean vector $\mu = [\mu_1 \ \mu_2]^T$. Also, identify the lengths of major and minor axes of this ellipse.

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ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Solve the following questions.

(4x5=20)

Q2. a) In common factor analysis, which standards do you consider while retaining the number of common factors. (5)

b) Briefly explain discrimination and classification. (5)

For (c) and (d), consider the details below of the sample from bivariate normal distribution.

$$n = 42, \bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}, S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

c) Test the hypothesis $H_0: \mu = [0.562 \ 0.589]$. (5)

d) Find 95% confidence ellipse for the population mean vector μ . (5)

Answer the following questions.

(2x15=30)

Q3. Let X_1 and X_2 be two random variables with covariance matrix: (15)

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

Obtain the principal components and find the percentage of variation explained by each principal component.

Q4. The covariance matrix for four standardized variables Z_1, Z_2, Z_3, Z_4 is, (15)

$$\rho = \begin{bmatrix} 1 & 0.4 & 0.5 & 0.6 \\ & 1 & 0.3 & 0.4 \\ & & 1 & 0.2 \\ & & & 1 \end{bmatrix}$$

Let $z_1 = [Z_1 \ Z_2]^T$ and $z_2 = [Z_3 \ Z_4]^T$. Obtain first pair of canonical variates and canonical correlation.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Multivariate Analysis

Course Code: STAT-414

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions. (6x5=30)

- i- Differentiate between confidence region and confidence interval.
- ii- What are the properties of principal components?
- iii- How communalities and loadings differ in common factor analysis?
- iv- Compare T^2 intervals and Bonferroni intervals.
- v- What are the criteria of retaining the number of factors in common factor analysis?
- vi- How would you define canonical variates and canonical correlations?

Write detailed answers to the following questions. (3x10=30)

- Q2- Consider the data below from two bivariate normal populations P_1 and P_2 with common covariance matrices.

$$X_1 = \begin{bmatrix} 2 & 12 \\ 4 & 10 \\ 3 & 8 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 5 & 7 \\ 3 & 9 \\ 4 & 5 \end{bmatrix}$$

Obtain linear discriminant function and allocate the new observation $x_0 = \begin{bmatrix} 1 \\ 4.4 \end{bmatrix}$ to one of the two populations.

- Q3- The eigen-values and eigen-vectors of a correlation matrix are given below.

$$\begin{aligned} \lambda_1 &= 2.85, e_1 = [0.33 \ 0.46 \ 0.38 \ 0.56 \ 0.47]^T \\ \lambda_2 &= 1.81, e_2 = [-0.61 \ 0.39 \ -0.56 \ 0.08 \ 0.40]^T \\ \lambda_3 &= 0.20, e_3 = [0.10 \ 0.74 \ 0.17 \ -0.60 \ -0.22]^T \\ \lambda_4 &= 0.10, e_4 = [0.14 \ -0.28 \ 0.12 \ -0.57 \ 0.75]^T \\ \lambda_5 &= 0.03, e_5 = [0.70 \ 0.07 \ -0.71 \ 0.00 \ 0.01]^T \end{aligned}$$

Assuming two-factor model, calculate and interpret common factor loadings, communalities and specific variances.

- Q4- Consider a sample taken from multivariate normal distribution.

$$n = 42, \bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}, S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

Find 95% confidence ellipse for the population mean vector. Also, calculate the shadows of this confidence ellipse on the axes.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions. (6x5=30)

- What is the difference between rotated and unrotated common factors?
- What is scree plot?
- Define common factor loadings and communalities.
- Explain the objective of canonical correlation analysis.
- Elaborate principal component analysis as a data reduction tool.
- Differentiate between simultaneous confidence intervals and confidence region.

Write detailed answers to the following questions. (3x10=30)

Q2. What are the properties of principal components? Derive the correlation between a variable and a principal component.

Q3. Measurements of stiffness and bending strength (in 100 pounds per inches square) for a sample of 7 pieces of a particular grade of lumber are given below. It is known that the sample comes from bivariate normal distribution. Obtain 95% confidence region for the population mean vector.

15 59.52 → 577.14 13.23 = C2

Stiffness (X_1)	12	11	22	18	19	16	15
Bending strength (X_2)	41	66	76	109	108	76	69

Q4. Consider the correlation matrix of four random variables Z_1, Z_2, Z_3, Z_4 . Partition these variables such that Z_1 and Z_2 are in the first part and Z_3 and Z_4 are in the second part. Compute canonical correlations and interpret them.

$$R = \begin{bmatrix} 1 & 0.4 & 0.5 & 0.6 \\ 0.4 & 1 & 0.3 & 0.4 \\ 0.5 & 0.3 & 1 & 0.2 \\ 0.6 & 0.4 & 0.2 & 1 \end{bmatrix}$$

$$\lambda_1 = 0.5490$$

$$\lambda_2 = 0.00093$$